William E. Eagan

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Active **SECRET** Security Clearance

PROFESSIONAL EXPERIENCE

Senior Design Engineer (Composite Structures)

October 2017 – Present

Karman Space and Defense (formerly Systima Technologies, Inc.) – Mukilteo, WA

- CATIA 3D Modeling & Automation:
  - Established CATIA as the company's primary composites design platform, implementing licensing, infrastructure, design standards, modeling practices, and automated drawing templates.
  - Accumulated 20,000+ hours of expertise in advanced composite part design, modeling, and parametric surfacing.
  - Develop and maintain a comprehensive suite of automation macros, streamlining engineering workflows, ensuring design consistency, and reducing manual errors.
- Composite Part and Process Development (R&D and Build to Print):
  - Lead designer for the High-Temperature Materials team, overseeing both one-off and production programs, including concept development, process definition, part design, process documentation, inspection plans, and the design of the associated tooling: layup molds, trim and drill fixtures, bonding jigs, etc.
  - Collaborate closely with production and manufacturing teams, guiding part and tool designs through the development cycle to ensure design success, adherence to customer requirements, on-time delivery, and successful FAI for production. Responsibilities include identifying potential issues or root causes of manufacturing defects and employing creative problem-solving to resolve them within project scope.
  - Conceptualize innovative techniques for producing a diverse range of composite parts, including aerodynamic fairings, carbon fiber monocoque structural components, extreme environment shingled ablatives, carbon-carbon composites, large-scale direct composite tooling solutions, supporting both space and military applications.
  - Collaborate with the engineering team to develop manufacturing process specifications and standards.
- Reverse Engineering and Spring-In Compensation
  - Developed deviation-field vector sampling algorithms to process scan data from as-built components and generate tooling compensation models for improved dimensional accuracy.

Mechanical Engineer/Composite Tool Designer

June 2015 – September 2017

Ascent Engineering (a subsidiary of Ascent Aerospace) – Bothell, WA

- Tooling Design: including concept development, modeling and drafting to both Boeing and Airbus standards.
- On Site*: Texstars LLC, Dallas, TX* (March – August, 2016) – Requalification effort for B2B Flash Protection Screens.
- On Site: *Boeing, Renton, WA* (Nov. 2016 – Apr. 2017) – Integrated member of tooling design department.

SKILLS PROFILE

- Expert in Windows Environment and Complete Composites Engineering Software Stack
  - CATIA V5-6 (parametric surfacing, composites workbench, reverse engineering, backend infrastructure, automation); programming in .NET (primarily VB with strong familiarity in C#), including SQL database integration; 2D ply nesting; Virtek; Hexagon Metrology Software (SpatialAnalyzer, PC-DMIS); G-Code; Adobe Creative Suite
- Thermosetting Composite and Plastic Processing Expertise:
  - Prepreg Layup, Vacuum Bagging, Autoclave Cure, 5-Axis Routing, Wet Layups, VARTM, RTM, SQRTM, Hot Draping, Crush Core, Bulk Resin Infusion, Compression Molding, Winding, Tape Wrapping
- Knowledge of polymer characterization techniques, including materials intended for extreme-temperature applications.
- Experienced designing for and working within the capabilities and limitations of industrial manufacturing equipment:
  - Mills, Lathes, Waterjet, Welding, Autoclaves/Hydroclaves, Ovens, Heated Presses, High Temperature Furnaces, Cutting Tables (including drag knife and ultrasonic), Virtek Projection Systems
- Metrology and Inspection:
  - Collaborated with machining and quality teams to establish standardized metrology practices when utilizing laser trackers (Leica) and portable CMM systems (Romer Arms).
  - Designed tooling and assemblies optimized for metrology-assisted locating, improving alignment accuracy, inspection efficiency, and manufacturing repeatability.
  - Experienced in applying complex GD&T schemes and developing inspection strategies for geometrically complex composite structures and molded surfaces.

EDUCATION

Western Washington University September 2010 – June 2014

Bachelor of Science in Plastics Engineering Technology (ABET Accredited)
Academic Minor: Materials Science

- Western Washington University President's List – 3.8 GPA – *Cum Laude*
- Treasurer and Student Outreach Officer - WWU SAMPE/SPE Chapter